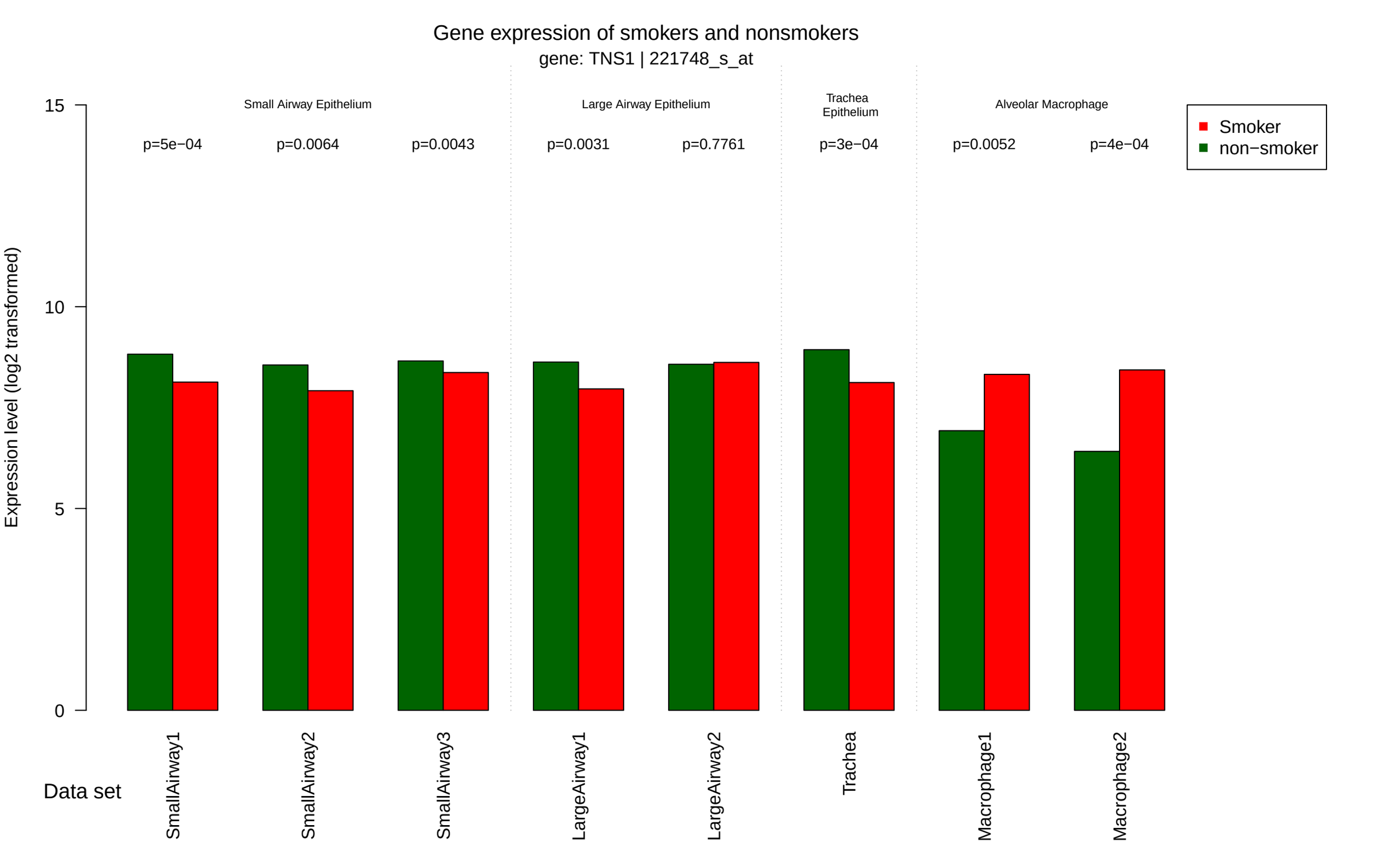
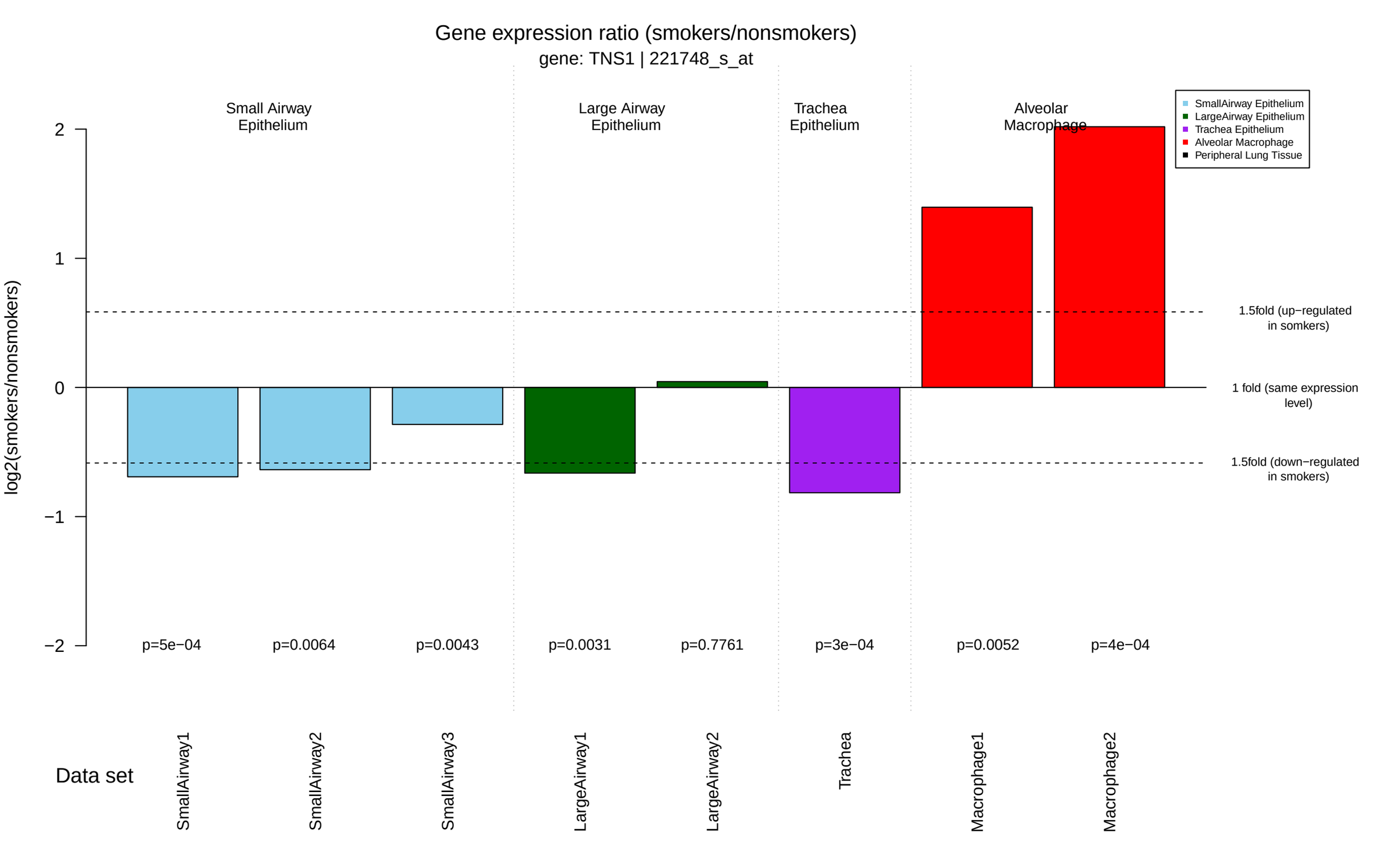
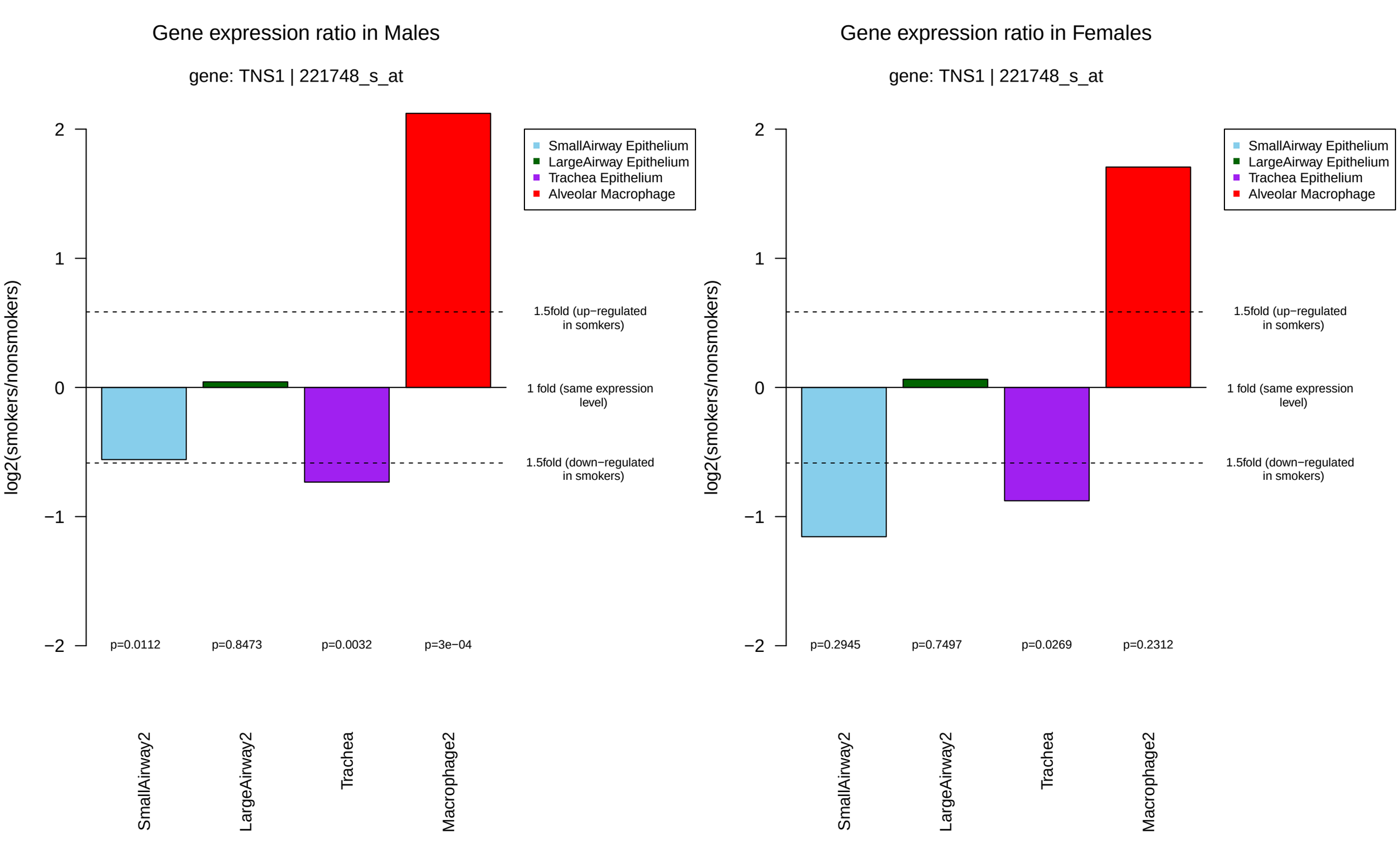
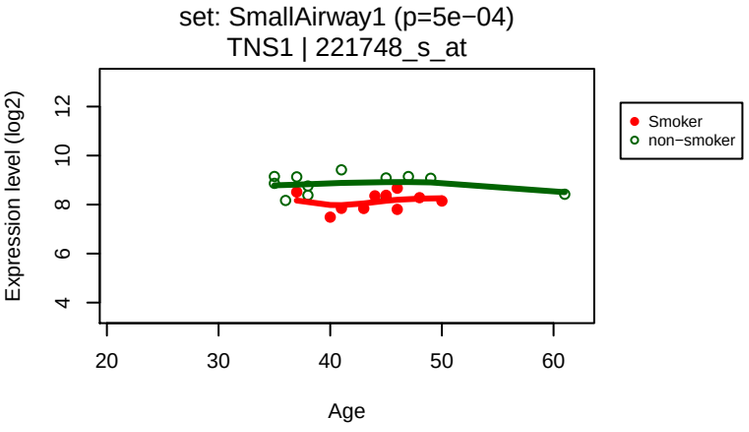
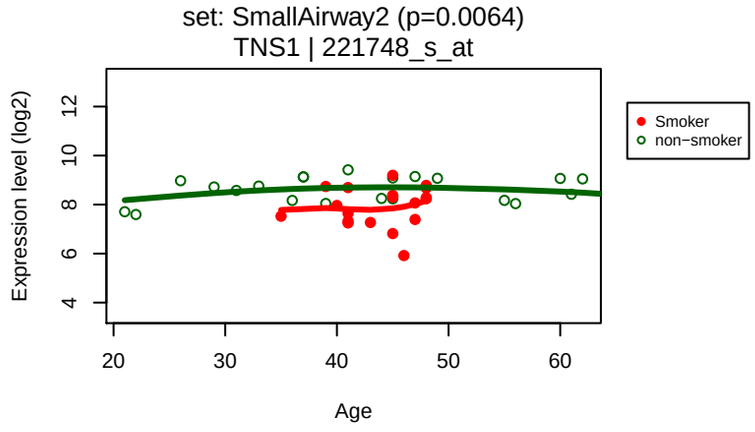
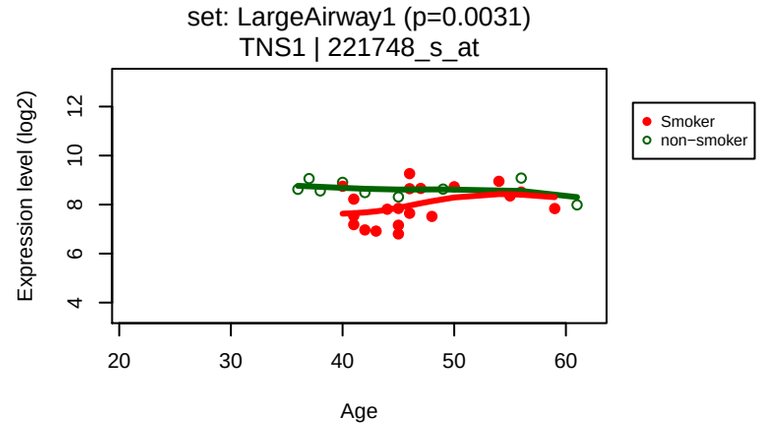

Gene expression of smokers and nonsmokers
gene: TNS1 | 221748_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=5e−04
p=0.0064
p=0.0043
p=0.0031
p=0.7761
p=3e−04
p=0.0052
p=4e−04
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

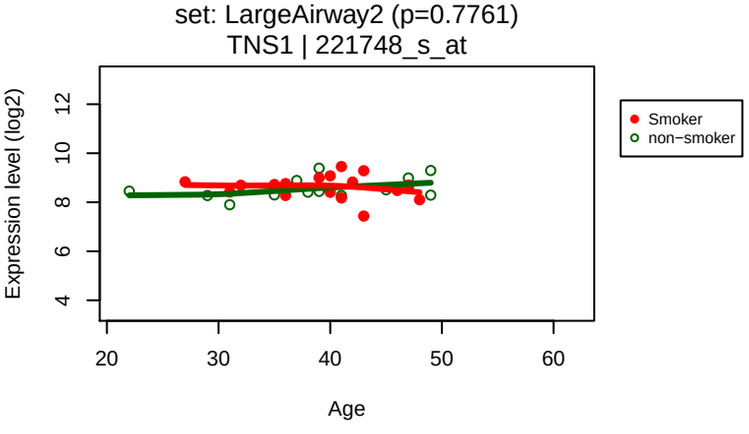

Gene expression ratio (smokers/nonsmokers)
gene: TNS1 | 221748_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=5e−04
p=0.0064
p=0.0043
p=0.0031
p=0.7761
p=3e−04
p=0.0052
p=4e−04
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

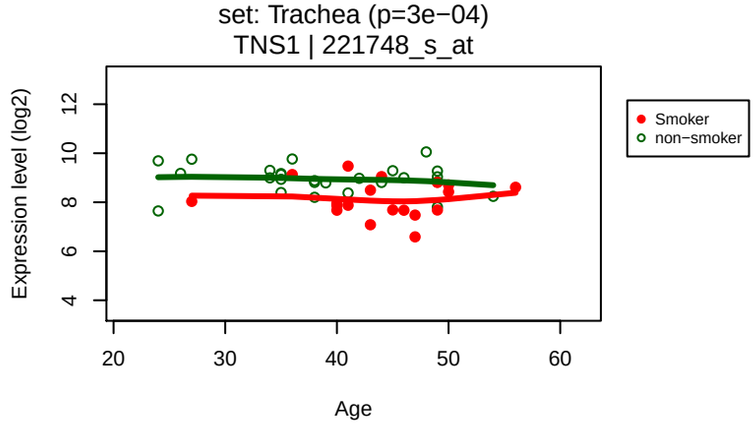

Gene expression ratio in Males
gene: TNS1 | 221748_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0112
p=0.8473
p=0.0032
p=3e−04
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TNS1 | 221748_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2945
p=0.7497
p=0.0269
p=0.2312
SmallAirway2
LargeAirway2
Trachea
Macrophage2

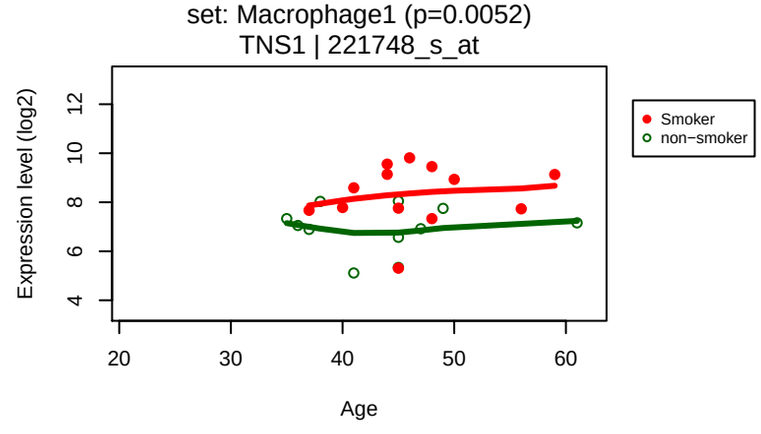

set: SmallAirway1 (p=5e−04)
TNS1 | 221748_s_at
Expression level (log2)
Age
Smoker
non−smoker

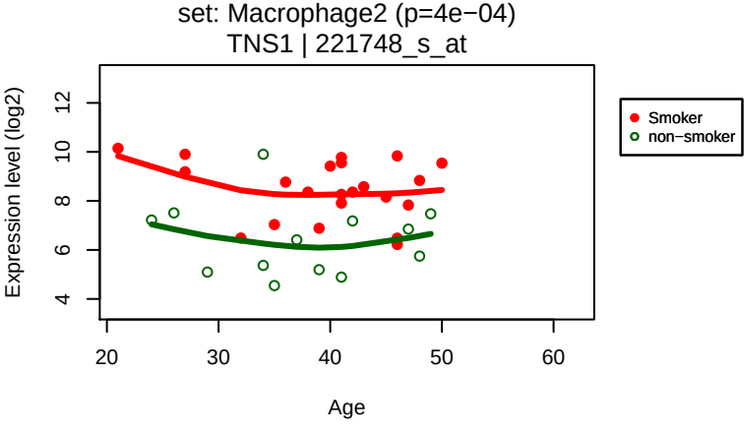

set: SmallAirway2 (p=0.0064)
TNS1 | 221748_s_at
Expression level (log2)
Age
Smoker
non−smoker

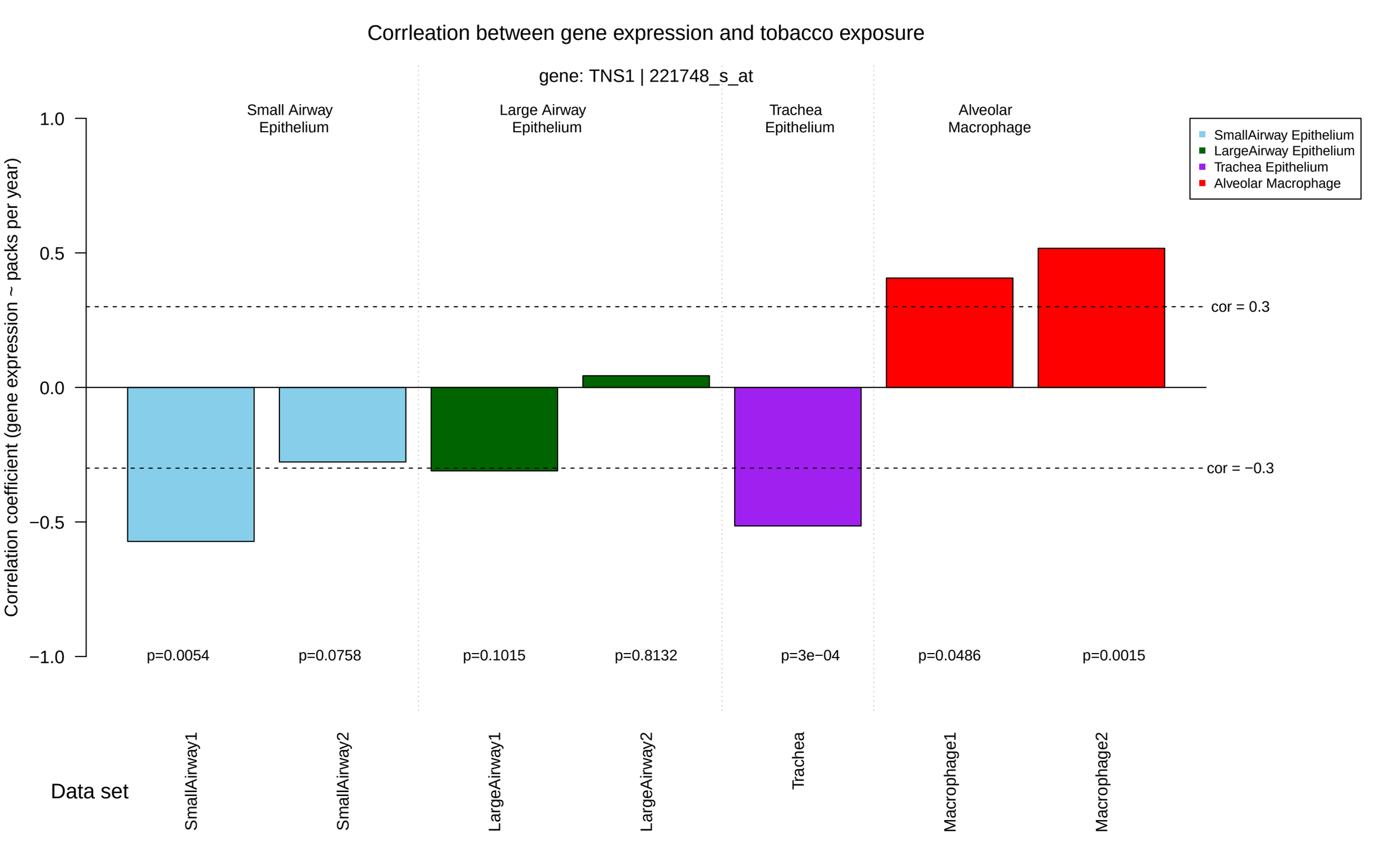

set: LargeAirway1 (p=0.0031)
TNS1 | 221748_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7761)
TNS1 | 221748_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=3e−04)
TNS1 | 221748_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0052)
TNS1 | 221748_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=4e−04)
TNS1 | 221748_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TNS1 | 221748_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0054
p=0.0758
p=0.1015
p=0.8132
p=3e−04
p=0.0486
p=0.0015
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2